

Human-AI Alignment

Alignment Specification

- Alignment Objectives
- Values & Stakeholders

Task & Assistance Alignment	FLAN (Wei et al., 2022); InstructGPT (Ouyang et al., 2022); WebGPT (Nakano et al., 2021).
Safety Alignment	Constitutional AI (Bai et al., 2022); Safe RLHF (Dai et al., 2024); Rule-Based Rewards (Mu et al., 2024).
Personalized Alignment	Personalizing RLHF (Poddar et al., 2024); Personalized Soups (Jang et al., 2023); Persona-Judge (Zhang et al., 2025).
Pluralistic & Societal Alignment	Collective Constitutional AI (Huang et al., 2024); PRISM (Kirk et al., 2024); Pluralistic Alignment (Sorensen et al., 2024).
Uncertainty & Drift	Cooperative IRL (Hadfield-Menell et al., 2016); Inverse Reward Design (Hadfield-Menell et al., 2017); COPR (Zhang et al., 2025).

Alignment Supervision

- Feedback Source
- Feedback & Oversight

Human Feedback	Deep RL from Human Preferences (Christiano et al., 2017); Learning to Summarize (Stiennon et al., 2020); RLHF-V (Yu et al., 2024).
AI Feedback	Constitutional AI (Bai et al., 2022); RLAIF (Lee et al., 2024); VLFeedback (Li et al., 2024).
Programmatic & Verifiable Feedback	Rule-Based Rewards (Mu et al., 2024); Math-Shepherd (Wang et al., 2024); DeepSeek-R1 (DeepSeek-AI, 2025).
Demonstrations & Preferences	Self-Instruct (Wang et al., 2023); UltraFeedback (Cui et al., 2024); HelpSteer2 (Wang et al., 2024).
Critique & Process Feedback	Fine-Grained Human Feedback (Wu et al., 2023); Let's Verify Step by Step (Lightman et al., 2024); RLHF-V (Yu et al., 2024).
Reliable & Scalable Oversight	Active Preference Learning (Muldrew et al., 2024); AI Safety via Debate (Irving et al., 2018); Weak-to-Strong (Burns et al., 2024).

Alignment Mechanisms

- Training-Time Alignment
- Inference-Time Alignment

Reward & Verifier Modeling	HelpSteer2 (Wang et al., 2024); Let's Verify Step by Step (Lightman et al., 2024); BiPRM (Zhang et al., 2026).
Supervised Alignment	FLAN (Wei et al., 2022); Self-Instruct (Wang et al., 2023); LIMA (Zhou et al., 2023).
Preference Optimization	DPO (Rafailov et al., 2023); TDPO (Zeng et al., 2024); SPPO (Wu et al., 2025).
Reinforcement Learning	InstructGPT (Ouyang et al., 2022); Safe RLHF (Dai et al., 2024); DeepSeek-R1 (DeepSeek-AI, 2025).
Alignment Distillation	PLaD (Zhang et al., 2024); PAD (Gu et al., 2025); AlignDistil (Zhang et al., 2025).
Steering, Search & Refinement	Inference-Time Intervention (Li et al., 2023); Self-Refine (Madaan et al., 2023); CRITIC (Gou et al., 2024).
Human Control	Corrigibility (Soares et al., 2015); Learning to Defer (Mozannar and Sontag, 2020); AI Control (Greenblatt et al., 2024).

Alignment Assurance

- Evaluation & Robustness
- System Assurance

Behavioral & Evaluator Evaluation	IFEval (Zhou et al., 2023); RewardBench (Lambert et al., 2025); PRMBench (Song et al., 2025).
Adversarial & Distribution Robustness	Universal Adversarial Attacks (Zou et al., 2023); HarmBench (Mazeika et al., 2024); Goal Misgeneralization in Deep RL (Di Langosco et al., 2022).
Alignment Preservation	Fine-Tuning Aligned LMs Compromises Safety (Qi et al., 2024); Lifelong Safety Alignment (Wang et al., 2025); LARF (Li et al., 2025).
Mechanistic & Theoretical Evidence	Discovering Latent Knowledge (Burns et al., 2023); Inference-Time Intervention (Li et al., 2023); Sustaining AI Safety (Mazzu, 2026).
Monitoring & Auditing	Model-Written Evaluations (Perez et al., 2023); AI Control (Greenblatt et al., 2024); Test-Set Contamination (Oren et al., 2024).